

ABILENE, TX, USA

WMO: 722660

Lat: 32.411N

Lon: 99.682W

Elev: 546

StdP: 94.94

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-4.2	-15.2	1.1	0.1	-12.8	1.3	2.9	13.3	11.6	12.2	11.9	4.4	0	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	38.1	21.6	36.9	21.6	35.7	21.7	24.4	32.2	23.9	31.6	23.4	30.9	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.3	26.8	21.9	17.7	26.4	21.3	17.1	26.2	76.6	32.0	74.3	31.5	72.2	30.8	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.7	9.3	DB	-10.0	40.1	2.1	1.6	-11.5	41.3	-12.8	42.2	-14.0	43.1	-15.5	44.3	(4)
			WB	-11.0	25.7	2.0	1.4	-12.4	26.7	-13.6	27.6	-14.7	28.4	-16.2	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	7.7	9.8	14.0	18.5	23.0	26.9	28.7	28.6	24.6	19.1	12.9	8.1
	DBStd	8.90	5.76	6.13	5.71	4.85	4.24	2.77	2.38	2.70	3.69	4.84	5.37	5.53
	HDD10.0	356	114	72	27	3	0	0	0	0	0	3	32	104
	HDD18.3	1341	331	243	154	57	11	0	0	0	3	51	172	319
	CDD10.0	3474	42	66	152	258	403	507	581	577	439	284	120	44
	CDD18.3	1417	1	4	21	61	155	257	323	319	192	73	11	1
	CDH23.3	16160	18	65	205	615	1617	2948	4063	3917	1975	638	84	15
	CDH26.7	8092	2	16	52	221	733	1499	2242	2173	926	213	13	2
Wind	WSAvg	4.9	4.8	5.1	5.5	5.9	5.4	5.2	4.4	4.0	4.0	4.7	4.7	4.6
Precipitation	PrecAvg	635	27	31	38	48	77	79	52	70	74	70	37	30
	PrecMax	1028	110	91	131	173	229	244	211	208	280	307	130	160
	PrecMin	352	0	0	1	0	11	0	0	0	0	3	0	0
	PrecStd	160	27	25	30	37	51	57	50	55	58	65	32	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.9	29.0	30.8	34.4	37.9	39.0	39.3	39.6	37.8	33.5	28.5	25.4
		MCWB	13.0	14.2	15.7	17.1	20.2	21.5	21.9	21.6	21.0	20.2	16.6	13.3
	2%	DB	22.2	25.2	28.0	31.5	35.0	36.8	37.7	37.9	35.3	31.1	25.8	21.9
		MCWB	11.9	14.0	15.7	17.4	20.6	21.7	21.8	21.7	21.3	19.6	16.3	12.5
	5%	DB	19.4	22.4	25.6	29.2	33.0	35.1	36.5	36.7	33.6	29.1	23.5	19.5
		MCWB	11.1	12.8	15.1	17.2	20.3	22.0	21.9	21.7	20.9	18.6	15.3	12.1
	10%	DB	16.9	19.4	23.1	27.0	30.8	33.4	35.3	35.3	31.8	27.1	21.3	17.0
		MCWB	10.0	11.7	14.1	16.8	20.3	22.1	22.1	21.8	20.8	17.9	14.3	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	17.9	19.5	21.5	24.0	25.4	25.2	24.6	24.0	22.8	19.9	17.5
		MCDB	20.6	22.1	25.4	27.9	31.4	33.3	33.4	32.4	31.0	29.2	24.8	20.5
	2%	WB	14.7	16.4	18.3	20.5	23.0	24.5	24.3	24.0	23.2	21.7	18.4	15.2
		MCDB	18.7	20.7	23.6	26.6	29.9	32.2	32.2	32.0	29.8	27.1	22.2	18.7
	5%	WB	12.6	14.9	17.3	19.7	22.3	23.7	23.7	23.4	22.6	20.7	17.0	13.3
		MCDB	17.1	19.8	22.1	25.1	29.1	31.2	31.4	31.5	29.0	25.9	21.1	17.6
	10%	WB	10.6	13.0	16.0	18.7	21.7	23.0	23.1	22.9	22.0	19.9	15.8	11.3
		MCDB	16.4	18.9	21.0	23.7	28.2	30.2	30.8	30.9	28.3	24.8	20.0	16.3
Mean Daily Temperature Range	5% DB	MDBR	12.8	12.9	12.9	13.3	12.3	11.7	11.8	11.8	11.9	12.3	12.3	12.2
		MCDBR	15.7	16.2	15.3	15.4	14.7	13.5	13.1	13.1	13.5	14.1	13.8	15.1
		MCWBR	8.5	8.7	7.4	7.2	5.4	4.3	3.9	3.9	4.3	6.2	7.1	8.5
	5% WB	MCDBR	13.3	13.9	12.1	12.5	12.4	11.7	11.5	11.5	11.1	11.3	11.5	12.0
		MCWBR	8.7	9.2	6.8	7.0	5.5	4.6	3.9	3.8	4.1	5.7	7.4	8.3
Clear-Sky Solar Irradiance	taub	0.289	0.303	0.324	0.367	0.391	0.406	0.416	0.404	0.374	0.350	0.317	0.296	
	taud	2.544	2.508	2.468	2.358	2.359	2.349	2.381	2.397	2.420	2.472	2.493	2.528	
	Ebn at Noon	942	959	958	922	897	879	868	876	893	892	897	912	
	Edh at Noon	82	94	106	124	125	126	121	117	110	97	86	80	
All-Sky Solar Radiation	RadAvg	3.19	3.85	5.05	6.13	6.44	7.08	6.94	6.33	5.30	4.25	3.36	2.81	
	RadStd	0.34	0.53	0.35	0.47	0.63	0.54	0.45	0.44	0.41	0.63	0.40	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	+89
Regional (18 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+84

Nomenclature: See separate page